



Will oil price surge again?

Impact of geopolitical events on crude oil prices - Abrar Alkhatlan

Project Description:

Oil prices are highly sensitive to geopolitical events that disrupt supply, threaten trade routes, and shift market expectations. This project examines how major geopolitical developments including conflicts and sanctions, impact global crude oil prices using 15 years of monthly data from 2010 to 2024. This analysis investigates whether and how conflict events correlate with price movements and quantifies the geopolitical risk premium markets price above supply fundamentals. The objective is to understand when and why political instability drives oil price volatility and when it does not.

Data Sources:

Source	Description	Period	Data Size
EIA (API)	Brent crude oil prices	2000-2024	289 monthly rows
EIA (API)	Oil Production	2000-2024	289 monthly rows
ACLED (CSV)	Political violence events	2010-2024	1.2M rows

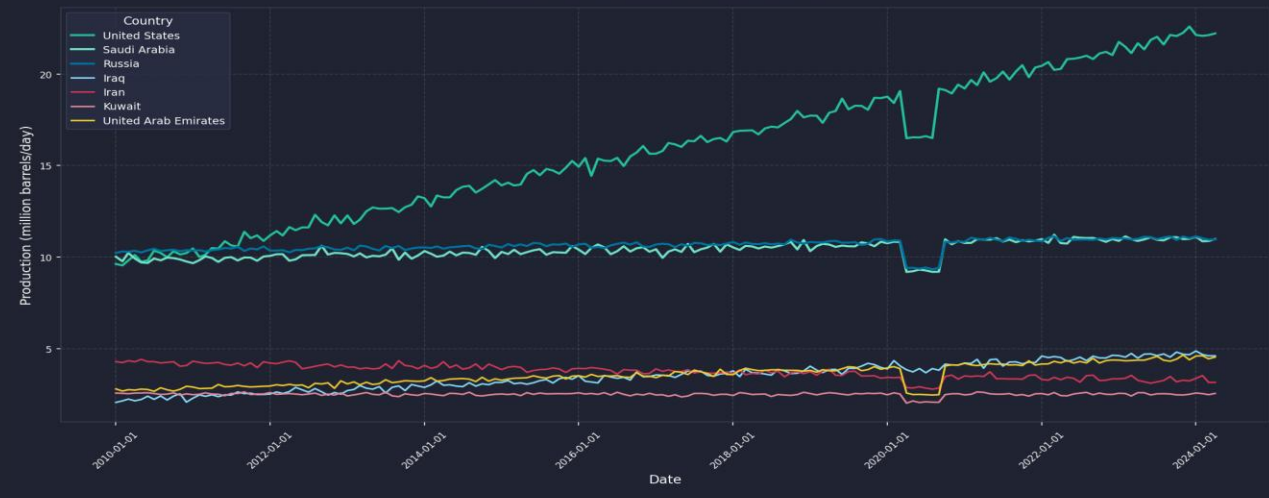
Before answering the question, I explored each data source to spot patterns

Monthly Oil Prices (USD per barrel)



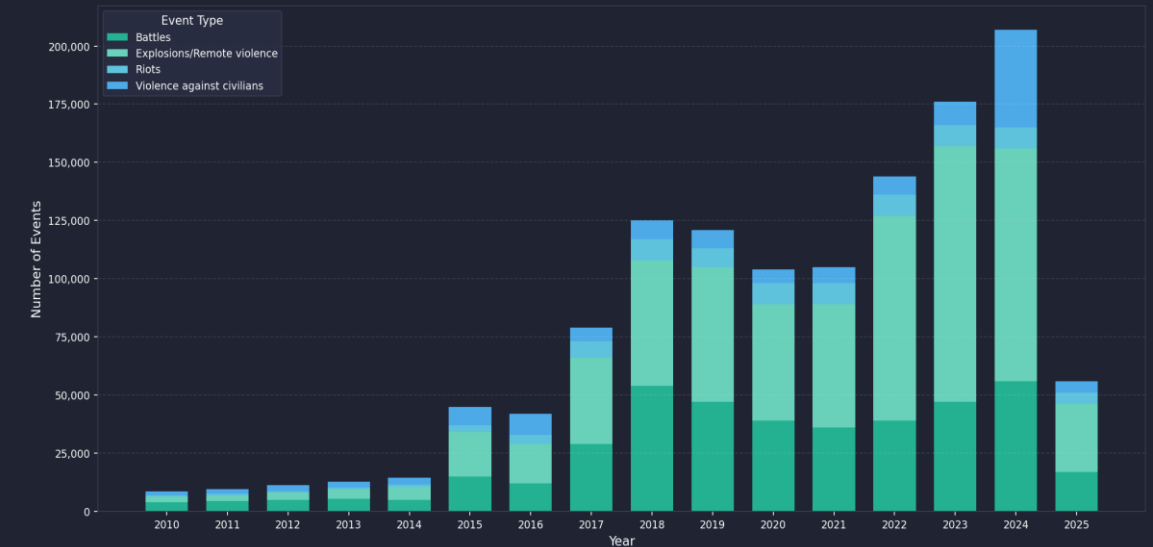
Oil prices spiked from \$18 at the COVID crash to \$125 during the Ukraine War

Monthly Oil Production by Country (mb/d)



The US grew from 4th to the largest oil producer over the last 15 years

Political Violence by Year (Event Type Breakdown)

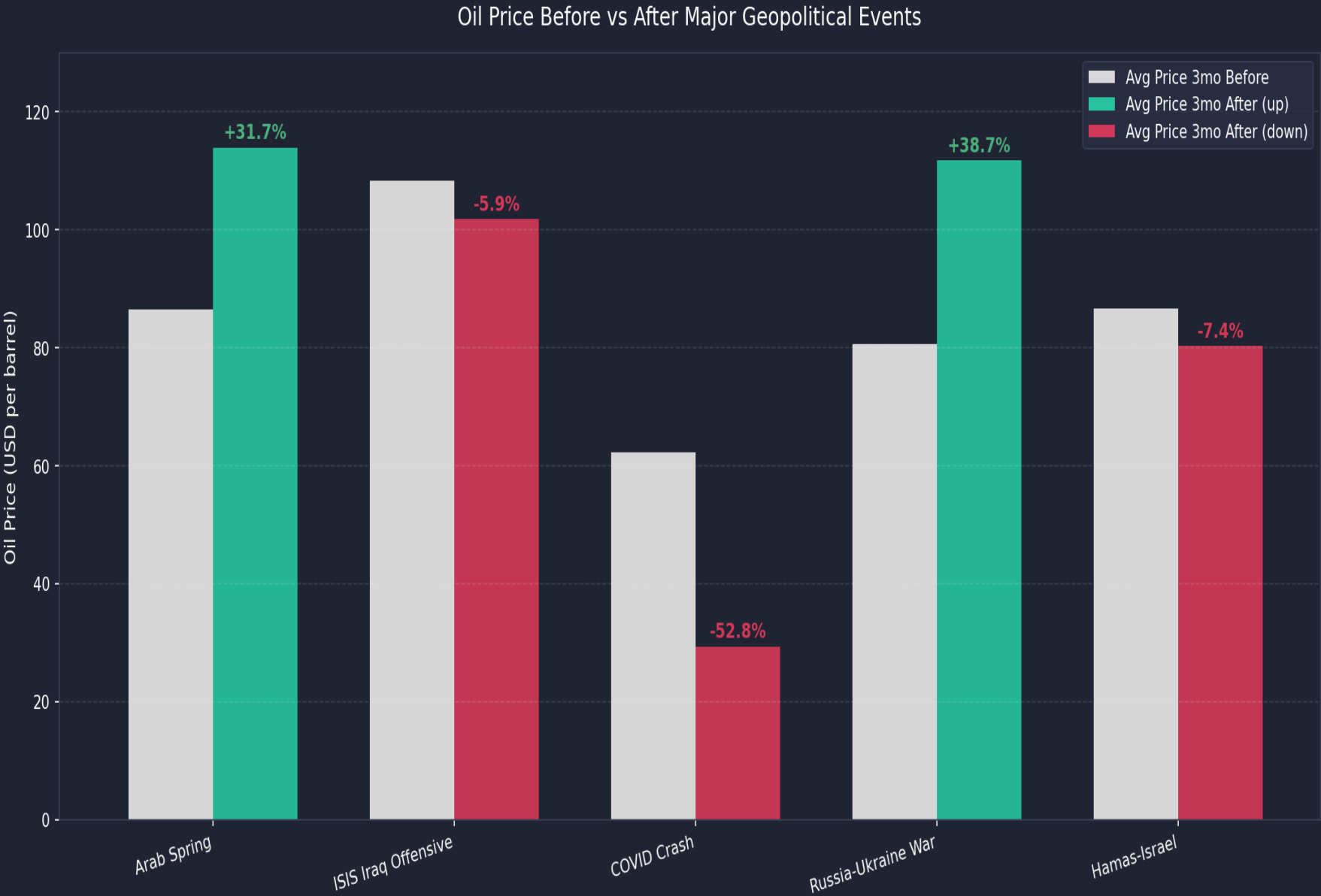


Global political violence grew tenfold, from 20,000 events in 2015 to over 200,000 in 2024. This raises the key question, if conflict grew tenfold, why didn't prices follow?

Oil Price vs Political Violence Over Time



Political violence was near zero from 2010 to 2015 while oil prices were at their highest above \$100. Then violence exploded from 2015 onward while prices collapsed and stayed low. The two lines move in opposite directions for most of the period



I compared prices 3 months before and after five major events. The pattern is clear:

When conflict stopped oil from reaching the market, prices went up:

- Arab Spring : Libyan production collapsed +31.7%
- Russia-Ukraine War : Russian exports sanctioned +38.7%

When conflict stayed contained and oil kept flowing, prices went down:

- Hamas-Israel: -7.4%
- ISIS Iraq: -5.9%
- COVID demand collapsed: -52.8%

Markets don't react to violence. They react to barrels

Fear Premium: how much extra does fear add to the price above what supply cost?

Actual Oil Price vs Supply Baseline
(Red = geopolitical premium, Blue = geopolitical discount)



Monthly Geopolitical Risk Premium in Oil Price



I built a baseline model using production data only, then measured the gap between actual price and that prediction.

- Ukraine War: +\$4.03/barrel above supply baseline
- COVID: -\$2.02/barrel below supply baseline
- Average across all 169 months: just \$0.56

For most of history supply fundamentals explain prices perfectly. Fear only adds up during real supply shocks. Always about the barrels, If oil keeps flowing the price stays flat or drops. If oil stops flowing the price spikes

Conclusion:

- Conflict type matters more than conflict volume: a tenfold increase in global violence did not cause a tenfold increase in prices
- Supply disruption is the only reliable price driver: when oil keeps flowing, price is unaffected regardless of conflict intensity
- Markets price geopolitical fear above baseline only during supply shocks, the average premium across 15 years is just \$0.56 per barrel

Challenges

Two main challenges:

- **Data Processing:** 1.2 million conflict records across 6 files, two live APIs, 15 years of daily data, all had to be cleaned and aligned before any analysis could start
- **Signal vs noise:** With so many factors affecting oil prices at the same time, figuring out which analysis actually answered my question and which ones were just noise required a lot of iteration, I went through multiple models and approaches before landing on a framework that could isolate the geopolitical impact



Thank you